

Notation

$\mathbb{Z}_n$	Set of all residue classes modulo n
$X \setminus Y$	The set of elements from X which are not in Y
$\mathbb{N}$	The set of all natural numbers $1, 2, 3, \dots$
$\mathbb{R}$	The set of all real numbers
S_n	Set of all permutations of the set $\{1, 2, \dots, n\}$
$GL_n(\mathbb{R})$	Set of all $n \times n$ invertible matrices with real entries
$\hat{i}, \hat{j}, \hat{k}$	unit vectors having the directions of the positive x , y and z axes in a three dimensional rectangular coordinate system, respectively
M^T	Transpose of a matrix M

SECTION – A
MULTIPLE CHOICE QUESTIONS (MCQ)

Q. 1 – Q.10 carry one mark each.

- Q.1 Consider the function $f(x, y) = 5 - 4 \sin x + y^2$ for $0 < x < 2\pi$ and $y \in \mathbb{R}$. The set of critical points of $f(x, y)$ consists of
- (A) a point of local maximum and a point of local minimum
(B) a point of local maximum and a saddle point
(C) a point of local maximum, a point of local minimum and a saddle point
(D) a point of local minimum and a saddle point

- Q.2 Let $\varphi: \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function such that φ' is strictly increasing with $\varphi'(1) = 0$. Let α and β denote the minimum and maximum values of $\varphi(x)$ on the interval $[2, 3]$, respectively. Then which one of the following is TRUE?

- (A) $\beta = \varphi(3)$ (B) $\alpha = \varphi(2.5)$ (C) $\beta = \varphi(2.5)$ (D) $\alpha = \varphi(3)$

- Q.3 The number of generators of the additive group $\mathbb{Z}_{36}$ is equal to

- (A) 6 (B) 12 (C) 18 (D) 36

Q.4

$$\lim_{n \rightarrow \infty} \frac{\pi}{n} \sum_{k=1}^n \sin\left(\frac{\pi}{2} + \frac{5\pi}{2} \cdot \frac{k}{n}\right) =$$

- (A) $\frac{2\pi}{5}$ (B) $\frac{5}{2}$ (C) $\frac{2}{5}$ (D) $\frac{5\pi}{2}$

- Q.5 Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a twice differentiable function. If $g(u, v) = f(u^2 - v^2)$, then

$$\frac{\partial^2 g}{\partial u^2} + \frac{\partial^2 g}{\partial v^2} =$$

- (A) $4(u^2 - v^2)f''(u^2 - v^2)$
(B) $4(u^2 + v^2)f''(u^2 - v^2)$
(C) $2f'(u^2 - v^2) + 4(u^2 - v^2)f''(u^2 - v^2)$
(D) $2(u - v)^2 f''(u^2 - v^2)$

Q.6

$$\int_0^1 \int_x^1 \sin(y^2) dy dx =$$

- (A) $\frac{1+\cos 1}{2}$ (B) $1 - \cos 1$ (C) $1 + \cos 1$ (D) $\frac{1-\cos 1}{2}$

Q.7 Let $f_1(x), f_2(x), g_1(x), g_2(x)$ be differentiable functions on $\mathbb{R}$. Let $F(x) = \begin{vmatrix} f_1(x) & f_2(x) \\ g_1(x) & g_2(x) \end{vmatrix}$ be the determinant of the matrix $\begin{bmatrix} f_1(x) & f_2(x) \\ g_1(x) & g_2(x) \end{bmatrix}$. Then $F'(x)$ is equal to

(A) $\begin{vmatrix} f_1'(x) & f_2'(x) \\ g_1(x) & g_2(x) \end{vmatrix} + \begin{vmatrix} f_1(x) & g_1'(x) \\ f_2(x) & g_2(x) \end{vmatrix}$

(B) $\begin{vmatrix} f_1'(x) & f_2'(x) \\ g_1(x) & g_2(x) \end{vmatrix} + \begin{vmatrix} f_1(x) & g_1'(x) \\ f_2(x) & g_2'(x) \end{vmatrix}$

(C) $\begin{vmatrix} f_1'(x) & f_2'(x) \\ g_1(x) & g_2(x) \end{vmatrix} - \begin{vmatrix} f_1(x) & g_1'(x) \\ f_2(x) & g_2'(x) \end{vmatrix}$

(D) $\begin{vmatrix} f_1'(x) & f_2'(x) \\ g_1'(x) & g_2'(x) \end{vmatrix}$

Q.8 Let

$$f(x) = \frac{x + |x|(1+x)}{x} \sin\left(\frac{1}{x}\right), \quad x \neq 0.$$

Write $L = \lim_{x \rightarrow 0^-} f(x)$ and $R = \lim_{x \rightarrow 0^+} f(x)$. Then which one of the following is TRUE?

(A) L exists but R does not exist

(B) L does not exist but R exists

(C) Both L and R exist

(D) Neither L nor R exists

Q.9 If $\lim_{T \rightarrow \infty} \int_0^T e^{-x^2} dx = \frac{\sqrt{\pi}}{2}$, then

$$\lim_{T \rightarrow \infty} \int_0^T x^2 e^{-x^2} dx =$$

(A) $\frac{\sqrt{\pi}}{4}$

(B) $\frac{\sqrt{\pi}}{2}$

(C) $\sqrt{2\pi}$

(D) $2\sqrt{\pi}$

Q.10 If

$$f(x) = \begin{cases} 1+x & \text{if } x < 0 \\ (1-x)(px+q) & \text{if } x \geq 0 \end{cases}$$

satisfies the assumptions of Rolle's theorem in the interval $[-1, 1]$, then the ordered pair (p, q) is

(A) $(2, -1)$

(B) $(-2, -1)$

(C) $(-2, 1)$

(D) $(2, 1)$

Q. 11 – Q. 30 carry two marks each.

Q.11 The flux of the vector field

$$\vec{F} = \left(2\pi x + \frac{2x^2 y^2}{\pi} \right) \hat{i} + \left(2\pi xy - \frac{4y}{\pi} \right) \hat{j}$$

along the outward normal, across the ellipse $x^2 + 16y^2 = 4$ is equal to

(A) $4\pi^2 - 2$

(B) $2\pi^2 - 4$

(C) $\pi^2 - 2$

(D) 2π

- Q.12 Let $\mathcal{M}$ be the set of all invertible 5×5 matrices with entries 0 and 1. For each $M \in \mathcal{M}$, let $n_1(M)$ and $n_0(M)$ denote the number of 1's and 0's in M , respectively. Then

$$\min_{M \in \mathcal{M}} |n_1(M) - n_0(M)| =$$

- (A) 1 (B) 3 (C) 5 (D) 15

- Q.13 Let $M = \begin{bmatrix} \frac{1}{2} & \frac{1}{4} \\ 0 & 1 \end{bmatrix}$ and $x = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$. Then

$$\lim_{n \rightarrow \infty} M^n x$$

- (A) does not exist (B) is $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$
 (C) is $\begin{bmatrix} 2 \\ 4 \end{bmatrix}$ (D) is $\begin{bmatrix} 3 \\ 4 \end{bmatrix}$

- Q.14 Let $\vec{F} = (3 + 2xy)\hat{i} + (x^2 - 3y^2)\hat{j}$ and let L be the curve

$$\vec{r}(t) = e^t \sin t \hat{i} + e^t \cos t \hat{j}, \quad 0 \leq t \leq \pi.$$

Then

$$\int_L \vec{F} \cdot d\vec{r} =$$

- (A) $e^{-3\pi} + 1$ (B) $e^{-6\pi} + 2$ (C) $e^{6\pi} + 2$ (D) $e^{3\pi} + 1$

- Q.15 The line integral of the vector field

$$\vec{F} = zx \hat{i} + xy \hat{j} + yz \hat{k}$$

along the boundary of the triangle with vertices $(1,0,0)$, $(0,1,0)$ and $(0,0,1)$, oriented anti-clockwise, when viewed from the point $(2,2,2)$, is

- (A) $\frac{-1}{2}$ (B) -2 (C) $\frac{1}{2}$ (D) 2

- Q.16 The area of the surface $z = \frac{xy}{3}$ intercepted by the cylinder $x^2 + y^2 \leq 16$ lies in the interval

- (A) $(20\pi, 22\pi]$ (B) $(22\pi, 24\pi]$ (C) $(24\pi, 26\pi]$ (D) $(26\pi, 28\pi]$

- Q.17 For $a > 0, b > 0$, let $\vec{F} = \frac{x\hat{j} - y\hat{i}}{b^2x^2 + a^2y^2}$ be a planar vector field. Let

$$C = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = a^2 + b^2\}$$

be the circle oriented anti-clockwise. Then

$$\oint_C \vec{F} \cdot d\vec{r} =$$

- (A) $\frac{2\pi}{ab}$ (B) 2π (C) $2\pi ab$ (D) 0
- Q.18 The flux of $\vec{F} = y\hat{i} - x\hat{j} + z^2\hat{k}$ along the outward normal, across the surface of the solid

$$\{(x, y, z) \in \mathbb{R}^3 \mid 0 \leq x \leq 1, 0 \leq y \leq 1, 0 \leq z \leq \sqrt{2 - x^2 - y^2}\}$$

is equal to

- (A) $\frac{2}{3}$ (B) $\frac{5}{3}$ (C) $\frac{8}{3}$ (D) $\frac{4}{3}$

- Q.19 Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function such that $f(2) = 2$ and

$$|f(x) - f(y)| \leq 5(|x - y|)^{3/2}$$

for all $x \in \mathbb{R}, y \in \mathbb{R}$. Let $g(x) = x^3 f(x)$. Then $g'(2) =$

- (A) 5 (B) $\frac{15}{2}$ (C) 12 (D) 24
- Q.20 Let $f: \mathbb{R} \rightarrow [0, \infty)$ be a continuous function. Then which one of the following is NOT TRUE?

- (A) There exists $x \in \mathbb{R}$ such that $f(x) = \frac{f(0) + f(1)}{2}$
 (B) There exists $x \in \mathbb{R}$ such that $f(x) = \sqrt{f(-1)f(1)}$
 (C) There exists $x \in \mathbb{R}$ such that $f(x) = \int_{-1}^1 f(t) dt$
 (D) There exists $x \in \mathbb{R}$ such that $f(x) = \int_0^1 f(t) dt$

Q.21 The interval of convergence of the power series

$$\sum_{n=1}^{\infty} \frac{1}{(-3)^{n+2}} \frac{(4x-12)^n}{n^2+1}$$

is

- (A) $\frac{10}{4} \leq x < \frac{14}{4}$
- (B) $\frac{9}{4} \leq x < \frac{15}{4}$
- (C) $\frac{10}{4} \leq x \leq \frac{14}{4}$
- (D) $\frac{9}{4} \leq x \leq \frac{15}{4}$

Q.22 Let $\mathcal{P}_3$ denote the real vector space of all polynomials with real coefficients of degree at most 3.

Consider the map $T: \mathcal{P}_3 \rightarrow \mathcal{P}_3$ given by $T(p(x)) = p''(x) + p(x)$. Then

- (A) T is neither one-one nor onto
- (B) T is both one-one and onto
- (C) T is one-one but not onto
- (D) T is onto but not one-one

Q.23 Let $f(x, y) = \frac{x^2 y}{x^2 + y^2}$ for $(x, y) \neq (0, 0)$. Then

- (A) $\frac{\partial f}{\partial x}$ and f are bounded
- (B) $\frac{\partial f}{\partial x}$ is bounded and f is unbounded
- (C) $\frac{\partial f}{\partial x}$ is unbounded and f is bounded
- (D) $\frac{\partial f}{\partial x}$ and f are unbounded

Q.24 Let S be an infinite subset of $\mathbb{R}$ such that $S \setminus \{\alpha\}$ is compact for some $\alpha \in S$. Then which one of the following is TRUE?

- (A) S is a connected set
- (B) S contains no limit points
- (C) S is a union of open intervals
- (D) Every sequence in S has a subsequence converging to an element in S

Q.25

$$\sum_{n=1}^{\infty} \tan^{-1} \frac{2}{n^2} =$$

- (A) $\frac{\pi}{4}$
- (B) $\frac{\pi}{2}$
- (C) $\frac{3\pi}{4}$
- (D) π

Q.26 Let $0 < a_1 < b_1$. For $n \geq 1$, define

$$a_{n+1} = \sqrt{a_n b_n} \text{ and } b_{n+1} = \frac{a_n + b_n}{2}.$$

Then which one of the following is NOT TRUE?

- (A) Both $\{a_n\}$ and $\{b_n\}$ converge, but the limits are not equal
- (B) Both $\{a_n\}$ and $\{b_n\}$ converge and the limits are equal
- (C) $\{b_n\}$ is a decreasing sequence
- (D) $\{a_n\}$ is an increasing sequence

Q.27

$$\lim_{n \rightarrow \infty} \frac{1}{\sqrt{n}} \left(\frac{1}{\sqrt{3} + \sqrt{6}} + \frac{1}{\sqrt{6} + \sqrt{9}} + \cdots + \frac{1}{\sqrt{3n} + \sqrt{3n+3}} \right) =$$

- (A) $1 + \sqrt{3}$ (B) $\sqrt{3}$ (C) $\frac{1}{\sqrt{3}}$ (D) $\frac{1}{1+\sqrt{3}}$

Q.28 Which one of the following is TRUE?

- (A) Every sequence that has a convergent subsequence is a Cauchy sequence
- (B) Every sequence that has a convergent subsequence is a bounded sequence
- (C) The sequence $\{\sin n\}$ has a convergent subsequence
- (D) The sequence $\left\{n \cos \frac{1}{n}\right\}$ has a convergent subsequence

Q.29 A particular integral of the differential equation

$$\frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} = e^{2x} \sin x$$

is

- (A) $\frac{e^{2x}}{10} (3 \cos x - 2 \sin x)$
- (B) $-\frac{e^{2x}}{10} (3 \cos x - 2 \sin x)$
- (C) $-\frac{e^{2x}}{5} (2 \cos x + \sin x)$
- (D) $\frac{e^{2x}}{5} (2 \cos x - \sin x)$

Q.30 Let $y(x)$ be the solution of the differential equation

$$(xy + y + e^{-x})dx + (x + e^{-x})dy = 0$$

satisfying $y(0) = 1$. Then $y(-1)$ is equal to

- (A) $\frac{e}{e-1}$ (B) $\frac{2e}{e-1}$ (C) $\frac{e}{1-e}$ (D) 0

SECTION - B

MULTIPLE SELECT QUESTIONS (MSQ)

Q. 31 – Q. 40 carry two marks each.

Q.31 For $\alpha, \beta \in \mathbb{R}$, define the map $\varphi_{\alpha, \beta}: \mathbb{R} \rightarrow \mathbb{R}$ by $\varphi_{\alpha, \beta}(x) = \alpha x + \beta$. Let

$$G = \{\varphi_{\alpha, \beta} \mid (\alpha, \beta) \in \mathbb{R}^2\}$$

For $f, g \in G$, define $g \circ f \in G$ by $(g \circ f)(x) = g(f(x))$. Then which of the following statements is/are TRUE?

- (A) The binary operation $\circ$ is associative
- (B) The binary operation $\circ$ is commutative
- (C) For every $(\alpha, \beta) \in \mathbb{R}^2$, $\alpha \neq 0$ there exists $(a, b) \in \mathbb{R}^2$ such that $\varphi_{\alpha, \beta} \circ \varphi_{a, b} = \varphi_{1, 0}$
- (D) $(G, \circ)$ is a group

Q.32 The volume of the solid

$$\left\{ (x, y, z) \in \mathbb{R}^3 \mid 1 \leq x \leq 2, \quad 0 \leq y \leq \frac{2}{x}, \quad 0 \leq z \leq x \right\}$$

is expressible as

- (A) $\int_1^2 \int_0^{2/x} \int_0^x dz dy dx$
- (B) $\int_1^2 \int_0^x \int_0^{2/x} dy dz dx$
- (C) $\int_0^2 \int_1^z \int_0^{2/x} dy dx dz$
- (D) $\int_0^2 \int_{\max\{z, 1\}}^2 \int_0^{2/x} dy dx dz$

Q.33 Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ be a function. Then which of the following statements is/are TRUE?

- (A) If f is differentiable at $(0,0)$, then all directional derivatives of f exist at $(0,0)$
- (B) If all directional derivatives of f exist at $(0,0)$, then f is differentiable at $(0,0)$
- (C) If all directional derivatives of f exist at $(0,0)$, then f is continuous at $(0,0)$
- (D) If the partial derivatives $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ exist and are continuous in a disc centered at $(0,0)$, then f is differentiable at $(0,0)$

Q.34 If X and Y are $n \times n$ matrices with real entries, then which of the following is/are TRUE?

- (A) If $P^{-1}XP$ is diagonal for some real invertible matrix P , then there exists a basis for $\mathbb{R}^n$ consisting of eigenvectors of X
- (B) If X is diagonal with distinct diagonal entries and $XY = YX$, then Y is also diagonal
- (C) If X^2 is diagonal, then X is diagonal
- (D) If X is diagonal and $XY = YX$ for all Y , then $X = \lambda I$ for some $\lambda \in \mathbb{R}$

- Q.35 Let G be a group of order 20 in which the conjugacy classes have sizes 1, 4, 5, 5, 5. Then which of the following is/are TRUE?
- (A) G contains a normal subgroup of order 5
- (B) G contains a non-normal subgroup of order 5
- (C) G contains a subgroup of order 10
- (D) G contains a normal subgroup of order 4
- Q.36 Let $\{x_n\}$ be a real sequence such that $7x_{n+1} = x_n^3 + 6$ for $n \geq 1$. Then which of the following statements is/are TRUE?
- (A) If $x_1 = \frac{1}{2}$, then $\{x_n\}$ converges to 1
- (B) If $x_1 = \frac{1}{2}$, then $\{x_n\}$ converges to 2
- (C) If $x_1 = \frac{3}{2}$, then $\{x_n\}$ converges to 1
- (D) If $x_1 = \frac{3}{2}$, then $\{x_n\}$ converges to -3
- Q.37 Let S be the set of all rational numbers in $(0,1)$. Then which of the following statements is / are TRUE?
- (A) S is a closed subset of $\mathbb{R}$
- (B) S is not a closed subset of $\mathbb{R}$
- (C) S is an open subset of $\mathbb{R}$
- (D) Every $x \in (0,1) \setminus S$ is a limit point of S
- Q.38 Let M be an $n \times n$ matrix with real entries such that $M^3 = I$. Suppose that $Mv \neq v$ for any non-zero vector v . Then which of the following statements is / are TRUE?
- (A) M has real eigenvalues
- (B) $M + M^{-1}$ has real eigenvalues
- (C) n is divisible by 2
- (D) n is divisible by 3
- Q.39 Let $y(x)$ be the solution of the differential equation

$$\frac{dy}{dx} = (y-1)(y-3)$$

satisfying the condition $y(0) = 2$. Then which of the following is/are TRUE?

- (A) The function $y(x)$ is not bounded above
- (B) The function $y(x)$ is bounded
- (C) $\lim_{x \rightarrow +\infty} y(x) = 1$
- (D) $\lim_{x \rightarrow -\infty} y(x) = 3$

Q.40 Let $k, \ell \in \mathbb{R}$ be such that every solution of

$$\frac{d^2 y}{dx^2} + 2k \frac{dy}{dx} + \ell y = 0$$

satisfies $\lim_{x \rightarrow \infty} y(x) = 0$. Then

- (A) $3k^2 + \ell < 0$ and $k > 0$
- (B) $k^2 + \ell > 0$ and $k < 0$
- (C) $k^2 - \ell \leq 0$ and $k > 0$
- (D) $k^2 - \ell > 0, k > 0$ and $\ell > 0$

SECTION - C

NUMERICAL ANSWER TYPE (NAT)

Q. 41 – Q. 50 carry one mark each.

Q.41 If the orthogonal trajectories of the family of ellipses $x^2 + 2y^2 = c_1$, $c_1 > 0$, are given by $y = c_2 x^\alpha$, $c_2 \in \mathbb{R}$, then $\alpha =$ _____

Q.42 Let G be a subgroup of $GL_2(\mathbb{R})$ generated by $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ and $\begin{bmatrix} 0 & -1 \\ 1 & -1 \end{bmatrix}$. Then the order of G is _____

Q.43 Consider the permutations $\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 4 & 5 & 3 & 7 & 8 & 6 & 1 & 2 \end{pmatrix}$ and $\tau = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 4 & 5 & 3 & 1 & 7 & 6 & 8 & 2 \end{pmatrix}$ in S_8 . The number of $\eta \in S_8$ such that $\eta^{-1} \sigma \eta = \tau$ is equal to _____

Q.44 Let P be the point on the surface $z = \sqrt{x^2 + y^2}$ closest to the point $(4, 2, 0)$. Then the square of the distance between the origin and P is _____

Q.45 $\left(\int_0^1 x^4 (1-x)^5 dx \right)^{-1} =$ _____

Q.46 Let $v_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ and $v_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$. Let M be the matrix whose columns are $v_1, v_2, 2v_1 - v_2, v_1 + 2v_2$ in that order. Then the number of linearly independent solutions of the homogeneous system of linear equations $Mx = 0$ is _____

Q.47
$$\frac{1}{2\pi} \left(\frac{\pi^3}{1!3} - \frac{\pi^5}{3!5} + \frac{\pi^7}{5!7} - \cdots + \frac{(-1)^{n-1} \pi^{2n+1}}{(2n-1)!(2n+1)} + \cdots \right) = \underline{\hspace{2cm}}$$

Q.48 Let P be a 7×7 matrix of rank 4 with real entries. Let $\mathbf{a} \in \mathbb{R}^7$ be a column vector. Then the rank of $P + \mathbf{a}\mathbf{a}^T$ is at least $\underline{\hspace{2cm}}$

Q.49 For $x > 0$, let $[x]$ denote the greatest integer less than or equal to x . Then
$$\lim_{x \rightarrow 0^+} x \left(\left\lfloor \frac{1}{x} \right\rfloor + \left\lfloor \frac{2}{x} \right\rfloor + \cdots + \left\lfloor \frac{10}{x} \right\rfloor \right) = \underline{\hspace{2cm}}$$

Q.50 The number of subgroups of $\mathbb{Z}_7 \times \mathbb{Z}_7$ of order 7 is $\underline{\hspace{2cm}}$

Q. 51 – Q. 60 carry two marks each.

Q.51 Let $y(x)$, $x > 0$ be the solution of the differential equation

$$x^2 \frac{d^2 y}{dx^2} + 5x \frac{dy}{dx} + 4y = 0$$

satisfying the conditions $y(1) = 1$ and $y'(1) = 0$. Then the value of $e^2 y(e)$ is $\underline{\hspace{2cm}}$

Q.52 Let T be the smallest positive real number such that the tangent to the helix

$$\cos t \hat{i} + \sin t \hat{j} + \frac{t}{\sqrt{2}} \hat{k}$$

at $t = T$ is orthogonal to the tangent at $t = 0$. Then the line integral of $\vec{F} = x\hat{j} - y\hat{i}$ along the section of the helix from $t = 0$ to $t = T$ is $\underline{\hspace{2cm}}$

Q.53 Let $f(x) = \frac{\sin \pi x}{\pi \sin x}$, $x \in (0, \pi)$, and let $x_0 \in (0, \pi)$ be such that $f'(x_0) = 0$. Then

$$(f(x_0))^2 (1 + (\pi^2 - 1) \sin^2 x_0) = \underline{\hspace{2cm}}$$

Q.54 The maximum order of a permutation σ in the symmetric group S_{10} is $\underline{\hspace{2cm}}$

Q.55 Let $a_n = \sqrt{n}$, $n \geq 1$, and let $s_n = a_1 + a_2 + \cdots + a_n$. Then

$$\lim_{n \rightarrow \infty} \left(\frac{a_n/s_n}{-\ln(1 - a_n/s_n)} \right) = \underline{\hspace{2cm}}$$

Q.56 For a real number x , define $[x]$ to be the smallest integer greater than or equal to x . Then

$$\int_0^1 \int_0^1 \int_0^1 ([x] + [y] + [z]) \, dx \, dy \, dz = \underline{\hspace{2cm}}$$

Q.57 For $x > 1$, let

$$f(x) = \int_1^x \left(\sqrt{\log t} - \frac{1}{2} \log \sqrt{t} \right) dt$$

The number of tangents to the curve $y = f(x)$ parallel to the line $x + y = 0$ is $\underline{\hspace{2cm}}$

Q.58 Let $\alpha, \beta, \gamma, \delta$ be the eigenvalues of the matrix

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{bmatrix}$$

Then $\alpha^2 + \beta^2 + \gamma^2 + \delta^2 = \underline{\hspace{2cm}}$

Q.59 The radius of convergence of the power series

$$\sum_{n=0}^{\infty} n! x^{n^2}$$

is $\underline{\hspace{2cm}}$

Q.60 If

$$y(x) = \int_{\sqrt{x}}^x \frac{e^t}{t} dt, \quad x > 0$$

then $y'(1) = \underline{\hspace{2cm}}$

END OF THE QUESTION PAPER

JAM 2017 ANSWER KEY

Model Answer Key for MA Paper

Paper: **MATHEMATICS**

Code: **MA**

SECTION – A (MCQ)				SECTION – B (MSQ)		SECTION – C (NAT Type)			
Q. No.	KEY	Q. No.	KEY	Q. No.	KEYS	Q. No.	KEY RANGE	Q. No.	KEY RANGE
01	D	16	A	31	A, C	41	1.9 – 2.1	56	2.9 – 3.1
02	A	17	A	32	A, B, D	42	5.9 – 6.1	57	0.9 – 1.1
03	B	18	D	33	A, D	43	-0.01 – +0.01	58	5.9 – 6.1
04	C	19	D	34	A, B, D	44	9.9 – 10.1	59	0.9 – 1.1
05	B	20	C	35	A, C	45	1259.9 – 1260.1	60	1.34 – 1.36
06	D	21	D	36	A, C	46	1.9 – 2.1		
07	B	22	B	37	B, D	47	0.49 – 0.51		
08	A	23	B	38	B, C	48	2.9 – 3.1		
09	A	24	D	39	B, C, D	49	54.9 – 55.1		
10	D	25	C	40	C, D	50	7.9 – 8.1		
11	B	26	A			51	2.9 – 3.1		
12	A	27	C			52	2.0 – 2.2		
13	C	28	C			53	0.9 – 1.1		
14	D	29	C			54	29.9 – 30.1		
15	C	30	B			55	0.9 – 1.1		